#Task 3

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Analyze Network Traffic by Using Wireshark, Nmap

Introduction

In this task, I used Wireshark to gain hands-on experience in monitoring and analyzing the data (traffic) that moves through the network while browsing the internet. The goal of this task is to help me learn how to observe the connection between my device and the websites I visit, and to understand the protocols used, such as TCP, UDP, DNS, and HTTP.

Through this experience, I started to understand how data moves, what it looks like, and how to tell the difference between normal packets and ones that may show a problem like delay or a dropped connection. I also learned how to use the Nmap tool to scan open ports on websites, which is helpful in checking network or website security.

This task helped me understand the basics of network analysis, and it’s an important first step in learning cybersecurity.

1) At the beginning, I downloaded the Wireshark program.

2) I have completed the Live Data Capture step, and while capturing, I generated data traffic by performing three of the required activities.

3) I found the following protocols during the packet capture, and here are some simple definitions about them.

- TCP (Transmission Control Protocol):
A connection-oriented protocol that ensures reliable and ordered delivery of data between devices. It checks for errors and guarantees that all data arrives intact.

- UDP (User Datagram Protocol):
A connectionless protocol that allows fast data transfer without guaranteeing delivery. It is commonly used in applications requiring speed, such as live streaming, DNS, and the QUIC protocol.

- DNS (Domain Name System):
This system translates human-readable domain names (like google.com) into IP addresses that computers use to communicate over the internet.

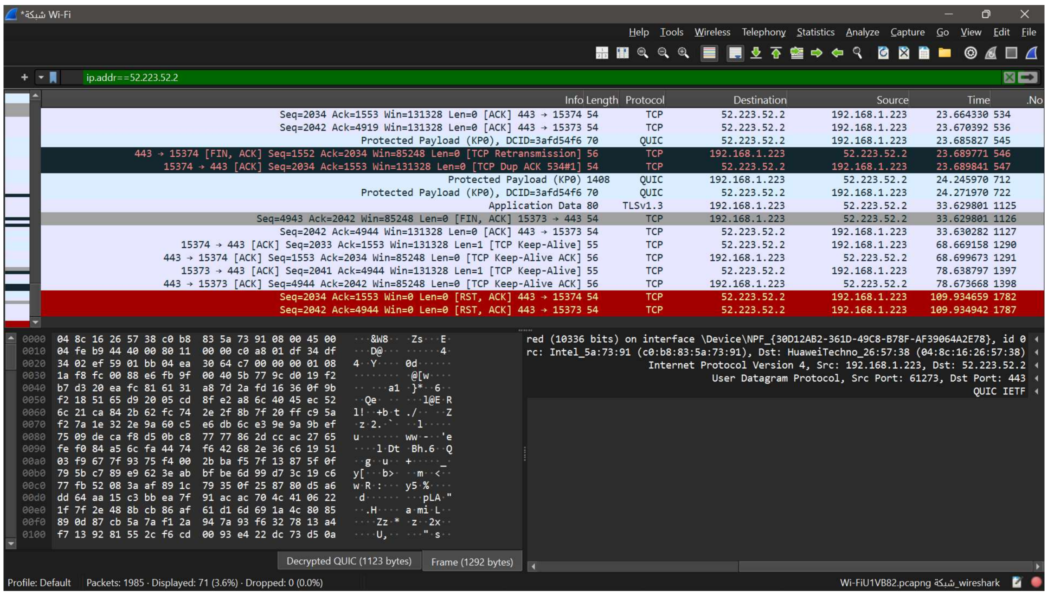
- ARP (Address Resolution Protocol):
Used to map IP addresses to physical MAC addresses within a local network, enabling devices to identify each other at the hardware level.

- HTTP (Hypertext Transfer Protocol):
The foundation of data communication on the web, used by browsers to request and receive web pages from servers.

- QUIC (Quick UDP Internet Connections):
A modern transport protocol built on UDP, designed to provide faster, more secure, and reliable connections for web browsing. Developed by Google.

4) Analyze One Selected Packet in screenshot and Detect and Analyze Packet Issues:

1-www.afaaqwetaan.com



--Analyze One Selected Packet:

- Packet Number: 534
- Protocol Type: TCP
- Source IP Address: 192.168.1.223
- Destination IP Address: 52.223.52.2
- Source Port: 15374
- Destination Port: 443

Description:
This packet is a TCP acknowledgment (ACK). It means the user’s device told the server that it got the data and is ready to receive more.

Technical Notes:
There was no delay or resend in this packet. This is normal in TCP communication to make sure data is sent correctly between the device and the server using HTTPS.

--Detect and Analyze Packet Issues :

- Number of affected packets: 1 selected (TCP Reset) other effected 22 red packets .
- Possible cause: Connection reset by the client, possibly due to timeout, app error, or network issue.
- Impact on connection: Connection with the server was abruptly terminated.
- Suggested solutions: Check internet stability, restart router, or try using a wired connection.

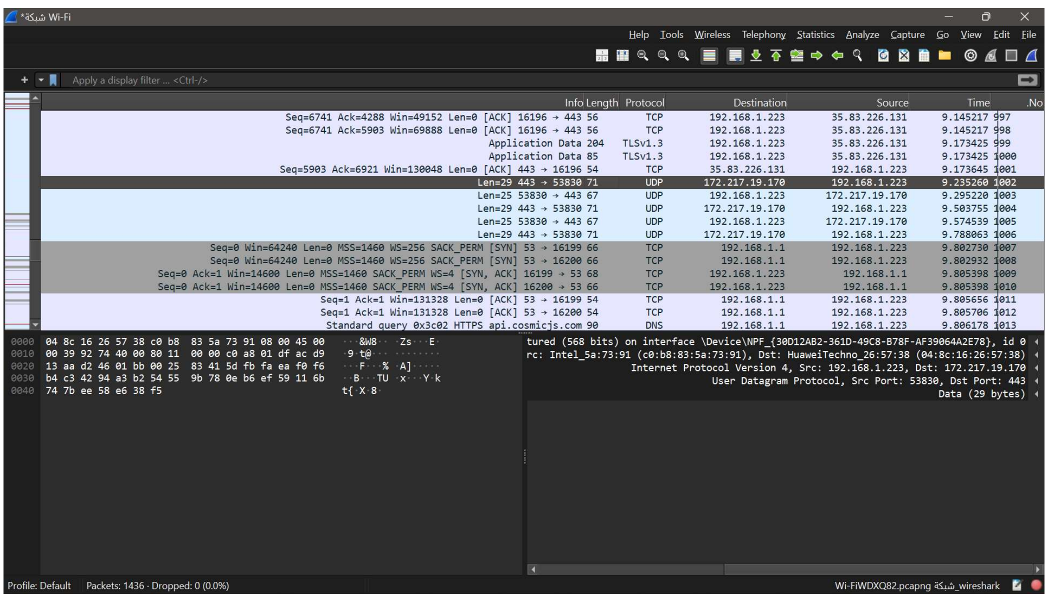
- • Wireshark Colors :

  Red – indicates a connection issue

  Red packets : 22

  Yellow packets : 44

2- www.wetaan.com



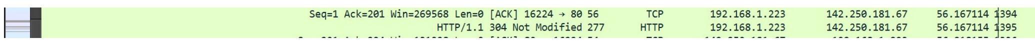
--Analyze One Selected Packet:

- Packet Number: 1002
- Protocol Type: UDP
- Source IP Address: 192.168.1.223
- Destination IP Address: 172.217.19.170
- Source Port: 53830
- Destination Port: 443

Description:
This is a small UDP packet sent from the device to a server. It is probably part of a QUIC connection which uses UDP for HTTPS on port 443.

Technical Notes:
There was no retransmission or delay in this packet. UDP does not guarantee delivery, so a higher protocol like QUIC handles making sure the data arrives correctly.

--Detect and Analyze Packet Issues :



- Packet Number: 1395
- Protocol Type: HTTP
- Number of affected packets: 1 selected (yellow), no retransmission observed.
- Possible cause: Cached response triggered yellow marking.
- Impact on connection: No negative effect; response was valid.
- Suggested solutions: No action needed – expected caching behavior.
- Wireshark Colors:
  Yellow/Orange = Warning (delay)

  Red packets : 10 effected

  Yellow packets :35 effected

3- https://mashaelalkhair.org.sa/

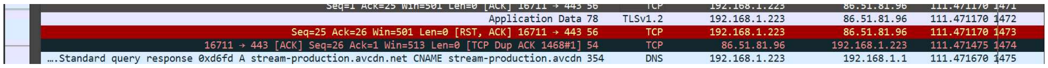
--Analyze One Selected Packet:

- Packet Number: 1475
- Protocol Type: DNS
- Source IP Address: 192.168.1.1
- Destination IP Address: 192.168.1.223
- Source Port: 53
- Destination Port: (Temporary port on device)

Description:
This packet is a DNS response for the website *stream-production.avcdn.net*. It shows the website's address by giving several IP addresses through something called a CNAME chain.

Technical Notes:
There are no problems with this packet. It's colored light blue, which means it is normal DNS traffic.

--Detect and Analyze Packet Issues:



- Packet Number: 1473
  - **Number of affected packets:** 1 (TCP Reset) red effected
  - **Possible cause:** Server closed the connection suddenly (timeout or rejection)
  - **Impact on connection:** Connection was interrupted
  - **Suggested solutions:** Check internet, reload page, restart router if needed
  - **Wireshark Colors :**

    Red – indicates a connection issue
- Red packets : 20 -Yellow packets: 28

- Sources:
  - Cisco:

    What is TCP?

    What is ARP?
  - Cloudflare:

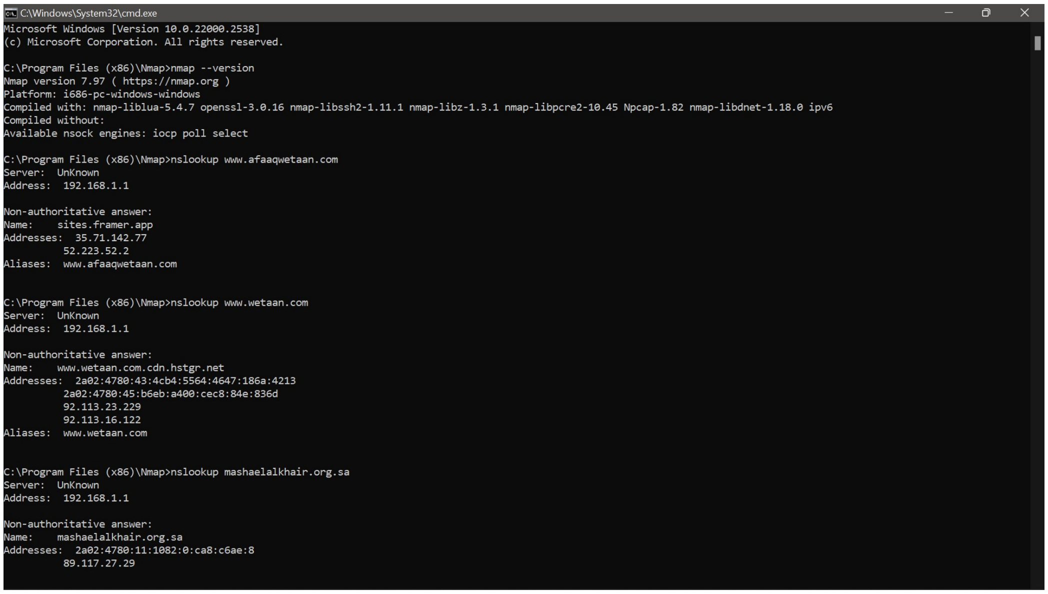
    What is UDP?

    What is DNS?

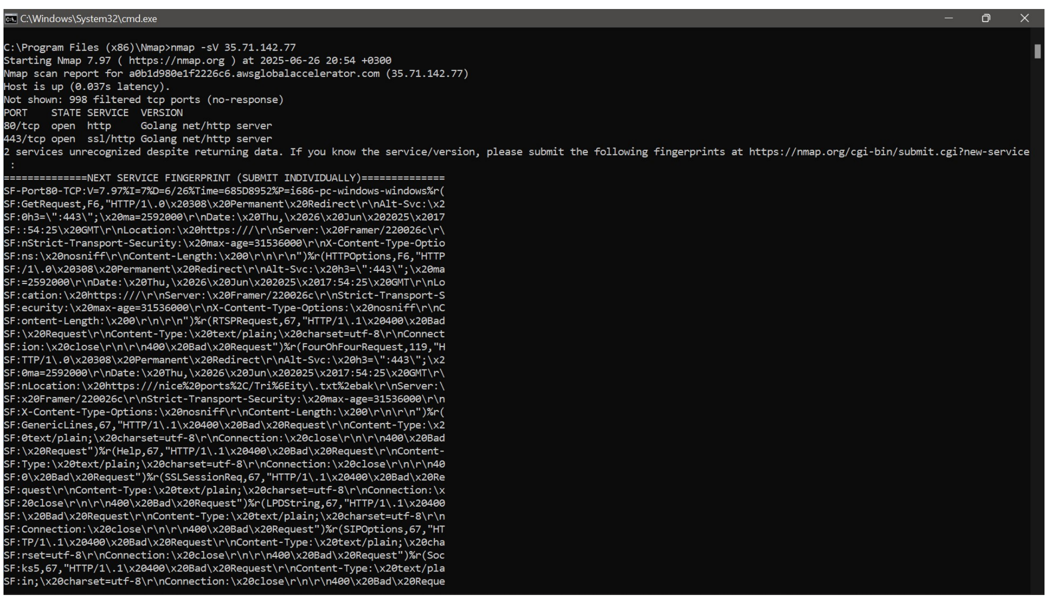
    What is QUIC?
  - MDN Web Docs:

    What is HTTP?

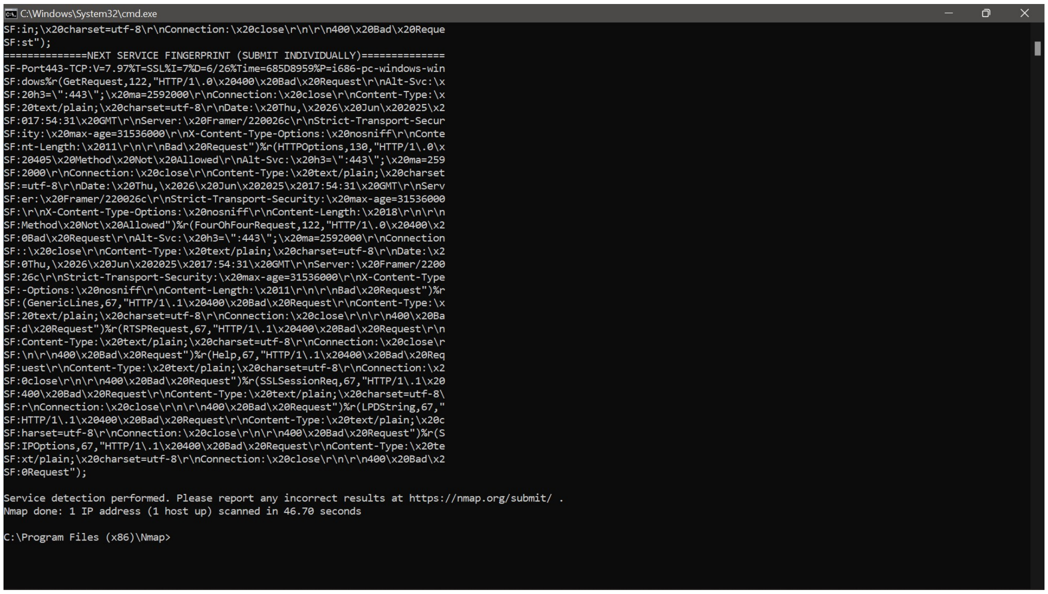
Part 2 :



I opened Command Prompt and navigated to the Nmap installation directory ,Then I ran the command nmap--version to check if the tool was properly installed. The terminal showed that Nmap version 7.97 is installed and ready to use.



<u>Port 1</u>



Port 2

We took the IP address and used Nmap to scan for open ports. Then, we identified the services running on those ports to understand what kind of connections the website allows. This helps us see if the open ports are normal for the site or if there might be potential security issues.

Port Scan and Service Analysis Using Nmap

Website 1: www.afaaqwetaan.com IP: 35.71.142.77

- Open Ports:

  80 (HTTP): Used for unencrypted web traffic, running Golang net/http server.

  443 (HTTPS): Used for encrypted web traffic, running Golang net/http server.

- Analysis:

  These ports are typical for a website.

  Presence of port 80 means unencrypted traffic is allowed, which may be a security concern.

  Recommended to enforce HTTPS and redirect all HTTP traffic to HTTPS.

```
C:\Program Files (x86)\Nmap>nslookup www.wetaan.com
Server:  UnKnown
Address:  192.168.1.1

Non-authoritative answer:
DNS request timed out.
    timeout was 2 seconds.
Name:    www.wetaan.com.cdn.hstgr.net
Addresses:  92.113.23.61
          92.113.16.10
Aliases:  www.wetaan.com


C:\Program Files (x86)\Nmap>nmap 92.113.23.61
Starting Nmap 7.97 ( https://nmap.org ) at 2025-06-26 21:17 +0300
Nmap scan report for 92.113.23.61
Host is up (0.096s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 8.87 seconds
```

Nmap Scan Report for www.wetaan.com   IP: 92.113.23.61

- Host Status: Host is up with 0.096s latency.
- Open Ports:

  80 tcp (HTTP): Unencrypted web traffic.

  443 tcp (HTTPS): Encrypted web traffic.

- Analysis:

  Both ports are typical and necessary for web servers.

  Port 80 allows unencrypted traffic, which may be a security risk if not redirected to HTTPS.

  No unusual or unexpected open ports detected.

```
C:\Program Files (x86)\Nmap>nmap 92.113.23.61
Starting Nmap 7.97 ( https://nmap.org ) at 2025-06-26 21:17 +0300
Nmap scan report for 92.113.23.61
Host is up (0.096s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 8.87 seconds

C:\Program Files (x86)\Nmap>nmap 89.117.27.29
Starting Nmap 7.97 ( https://nmap.org ) at 2025-06-26 21:29 +0300
Nmap scan report for 89.117.27.29
Host is up (0.20s latency).
Not shown: 962 filtered tcp ports (no-response), 34 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
80/tcp   open  http
443/tcp  open  https
3306/tcp open  mysql

Nmap done: 1 IP address (1 host up) scanned in 12.33 seconds

C:\Program Files (x86)\Nmap>
```

Nmap Scan Report for https://mashaelalkhair.org.sa/ IP: 89.117.27.29

- Host Status: Host is up (0.20s latency).
- Open Ports:

  21 tcp (FTP): File transfer service; may pose security risks if not secured.

  80 tcp (HTTP): Unencrypted web traffic.

  443 tcp (HTTPS): Encrypted web traffic.

3306 tcp (MySQL): Database service; should be protected from direct internet access.

- Analysis:

  Ports are typical for a web server with database support.

  FTP and MySQL open ports could be security concerns if not properly secured.

  Recommend restricting MySQL access and redirecting HTTP traffic to HTTPS.

Conclusion

In summary, this task provided valuable hands-on experience in understanding and analyzing network traffic using Wireshark to examine packet details and identify issues, and Nmap to scan open ports and assess security vulnerabilities. This experience enhanced my understanding of network security fundamentals and how to monitor them effectively.